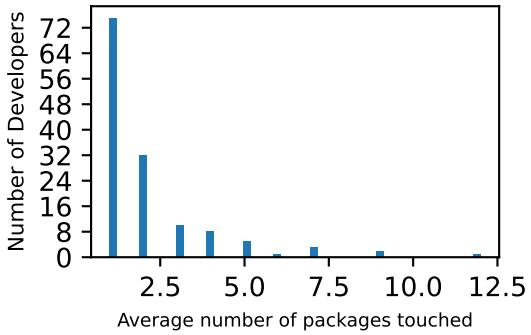
Learning a Codebase - Normal Distribution

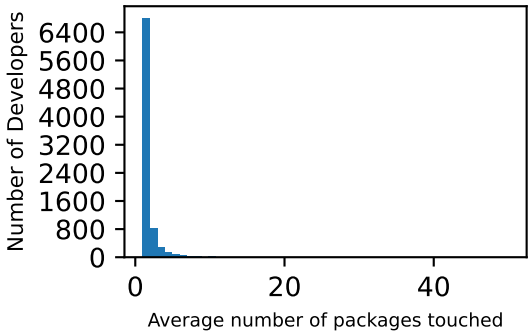
Derek Somerville

June 3, 2026

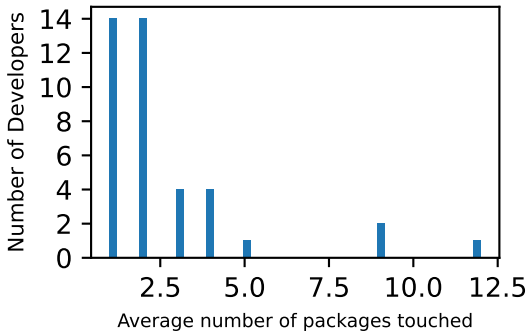
Contents



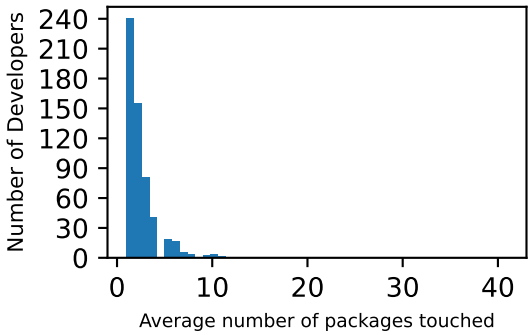
(a) Founder (μ: 2.07, median:1, mode:1, std:1.82 and max:12)



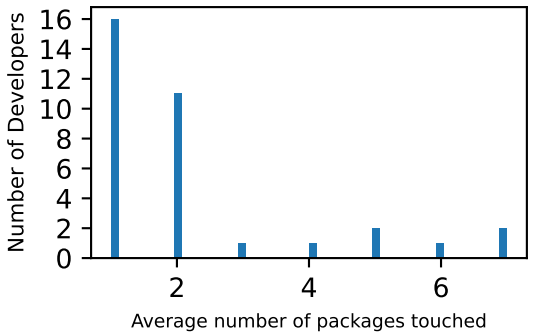
(b) Joiner (μ: 1.43, median:1, mode:1, std:1.62 and max:50)



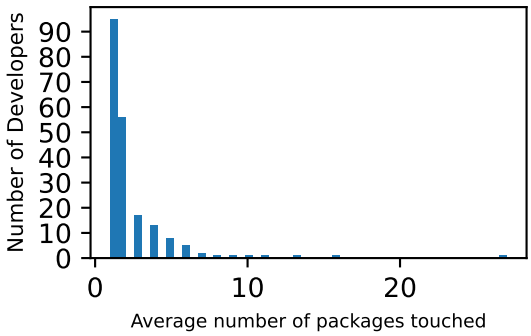
(c) Moderate founder (μ: 2.62, median:2.0, mode:1, std:2.4 and max:12)



(d) Moderate later joiner (μ: 2.55, median:2.0, mode:1, std:2.92 and max:41)

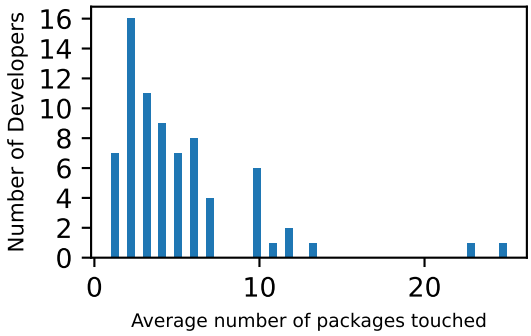


(e) Sustained founder (μ: 2.21, median:2.0, mode:1, std:1.77 and max:7)

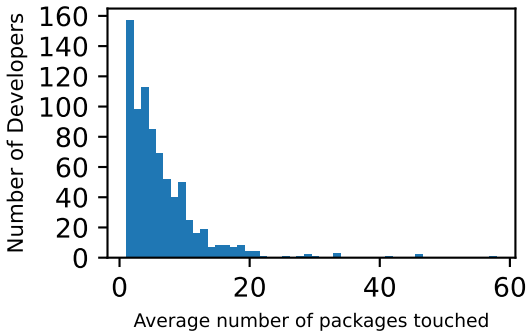


(f) Sustained later joiner (μ: 2.4, median:2, mode:1, std:2.73 and max:27)

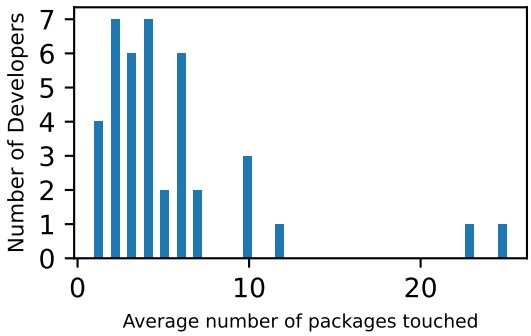
Figure 1: Packages for one commits for different categories of founder versus different categories of late joiner



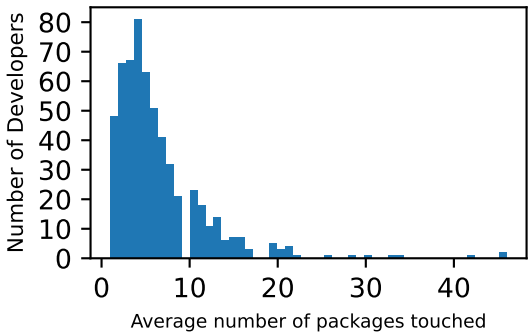
(a) Founder (μ: 5.07, median:4.0, mode:2, std:4.39 and max:25)



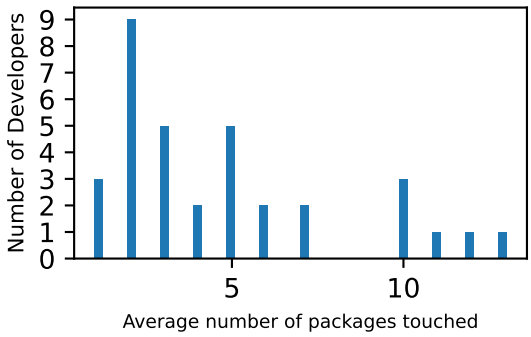
(b) Joiner (μ: 6.31, median:5, mode:4, std:5.71 and max:58)



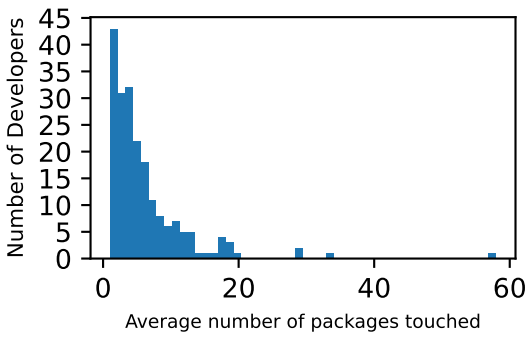
(c) Moderate founder (μ: 5.35, median:4.0, mode:2, std:5.1 and max:25)



(d) Moderate later joiner (μ: 6.38, median:5.0, mode:4, std:5.5 and max:46)

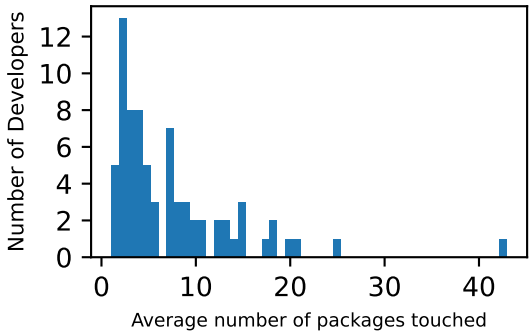


(e) Sustained founder (μ: 4.74, median:3.5, mode:2, std:3.41 and max:13)

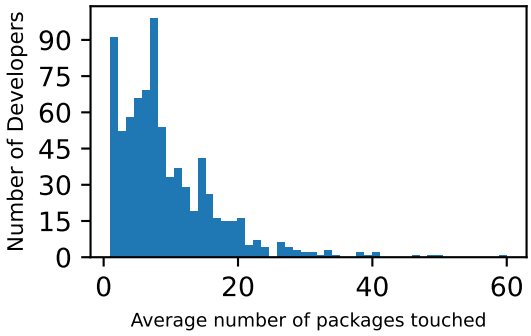


(f) Sustained later joiner (μ: 6.13, median:4, mode:4, std:6.31 and max:58)

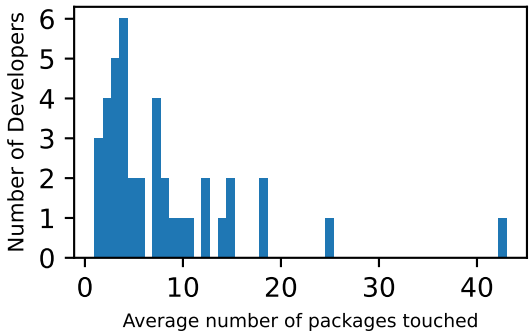
Figure 2: Packages for five commits for different categories of founder versus different categories of late joiner



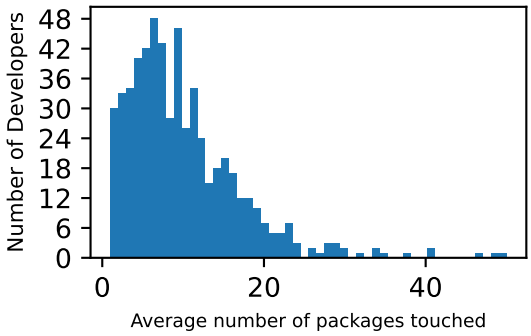
(a) Founder (μ: 7.34, median:5.0, mode:2, std:6.9 and max:43)



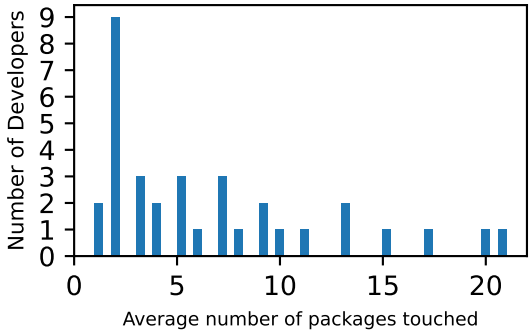
(b) Joiner (μ: 9.45, median:7, mode:6, std:7.47 and max:60)



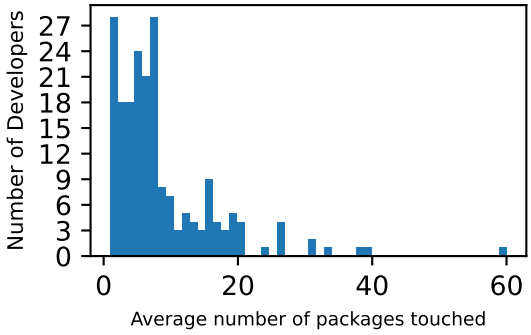
(c) Moderate founder (μ: 7.95, median:5.5, mode:4, std:7.9 and max:43)



(d) Moderate later joiner (μ: 9.73, median:8.0, mode:6, std:7.28 and max:50)

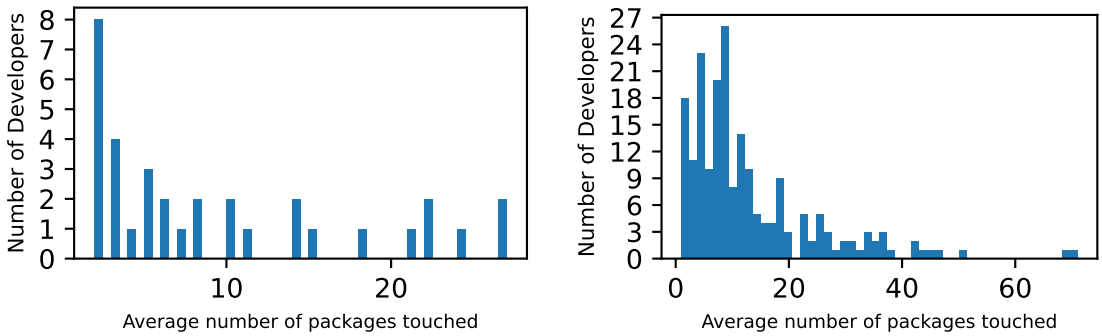


(e) Sustained founder (μ: 6.62, median:5.0, mode:2, std:5.52 and max:21)



(f) Sustained later joiner (μ: 8.65, median:6, mode:5, std:7.95 and max:60)

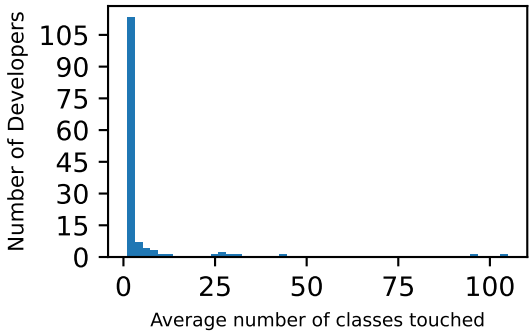
Figure 3: Packages for ten commits for different categories of founder versus different categories of late joiner



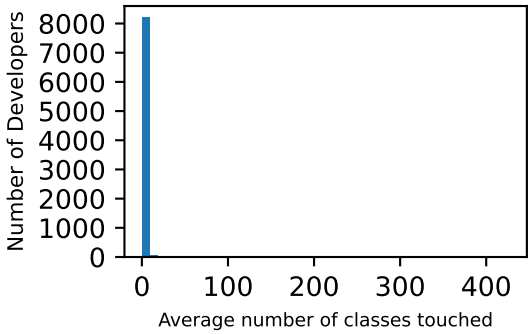
(a) Founder (μ: 9.32, median:6.0, mode:2, std:8.09 and max:27)

(b) Joiner (μ: 13.06, median:9, mode:7, std:11.89 and max:71)

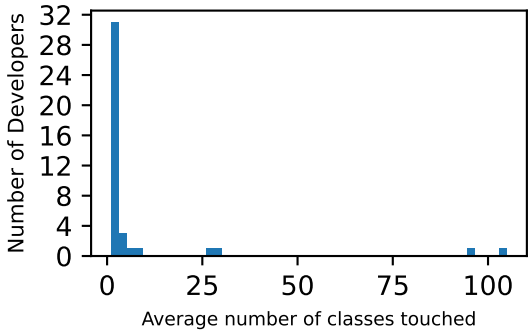
Figure 4: Packages for twenty commits for different categories of founder versus different categories of late joiner



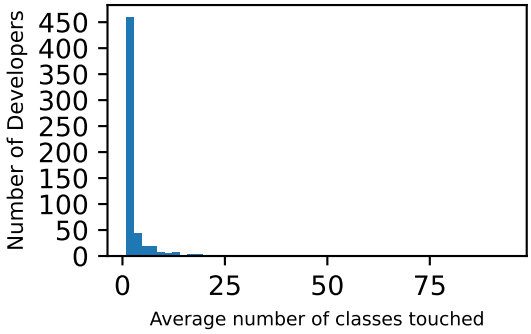
(a) Founder (μ: 4.59, median:1, mode:1, std:13.34 and max:105)



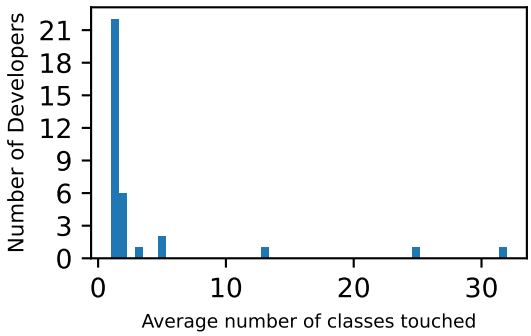
(b) Joiner (μ: 1.53, median:1, mode:1, std:6.04 and max:426)



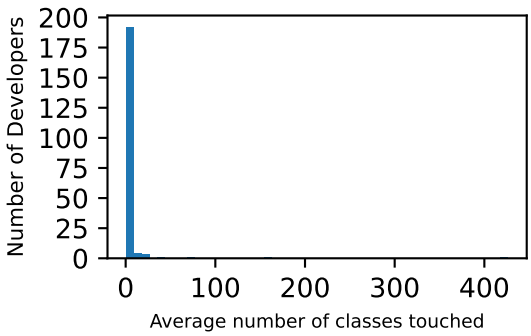
(c) Moderate founder (μ: 8.05, median:1.0, mode:1, std:22.27 and max:105)



(d) Moderate later joiner (μ: 2.98, median:1.0, mode:1, std:6.97 and max:94)

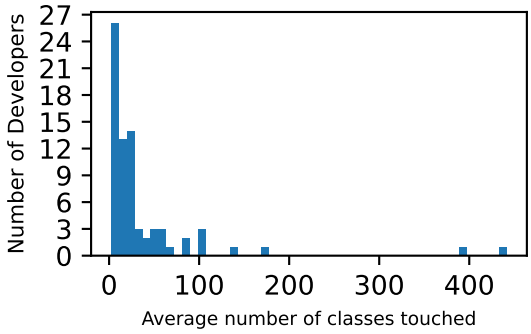


(e) Sustained founder (μ: 3.44, median:1.0, mode:1, std:6.79 and max:32)

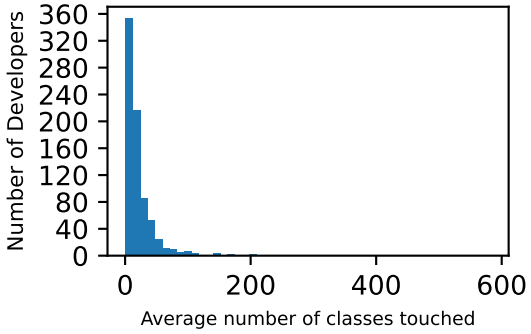


(f) Sustained later joiner (μ: 5.49, median:1, mode:1, std:32.41 and max:426)

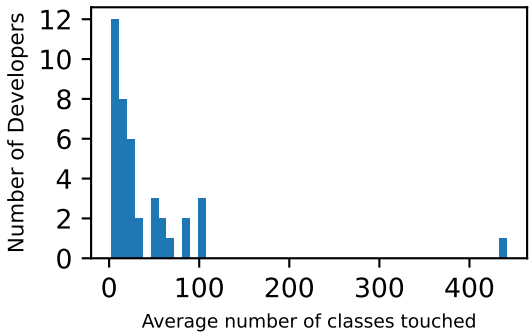
Figure 5: Classes for one commits for different categories of founder versus different categories of late joiner



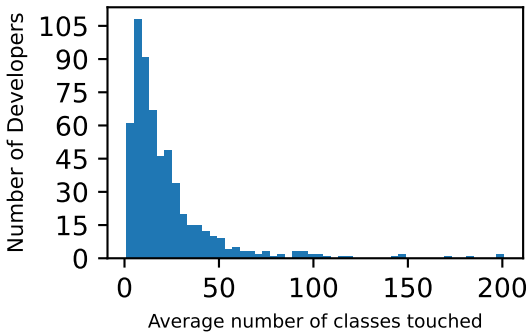
(a) Founder (μ: 39.05, median:18.5, mode:7, std:71.38 and max:442)



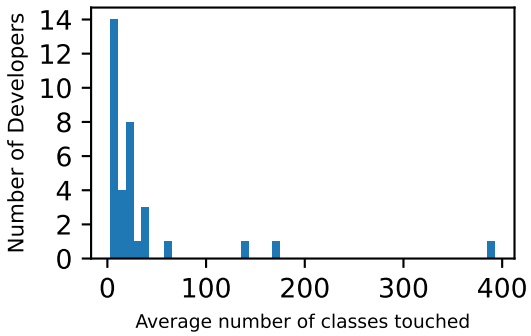
(b) Joiner (μ: 24.11, median:14, mode:6, std:40.17 and max:582)



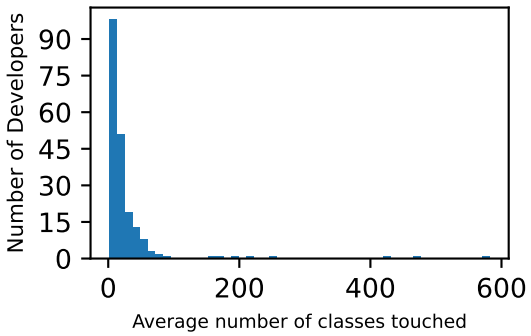
(c) Moderate founder (μ: 41.38, median:20.0, mode:5, std:71.59 and max:442)



(d) Moderate later joiner (μ: 22.04, median:14.0, mode:6, std:25.42 and max:201)

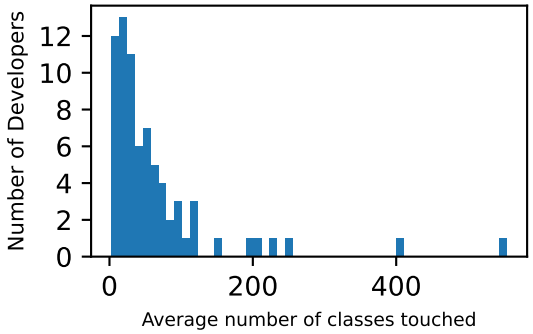


(e) Sustained founder (μ: 36.32, median:16.5, mode:7, std:72.12 and max:393)

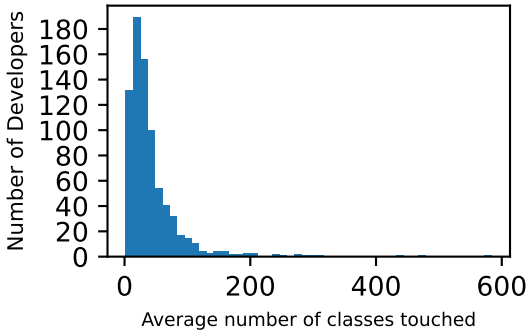


(f) Sustained later joiner (μ: 30.01, median:14, mode:9, std:65.92 and max:582)

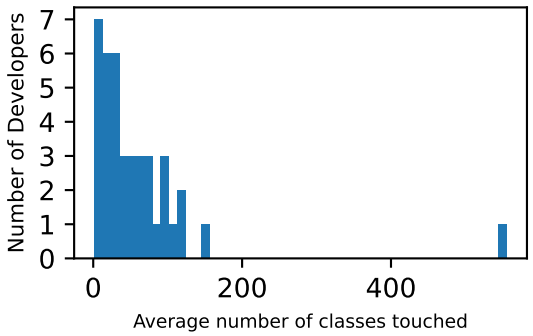
Figure 6: Classes for five commits for different categories of founder versus different categories of late joiner



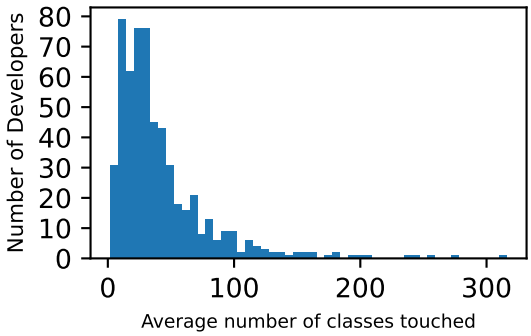
(a) Founder (μ: 64.81, median:38.0, mode:40, std:87.23 and max:555)



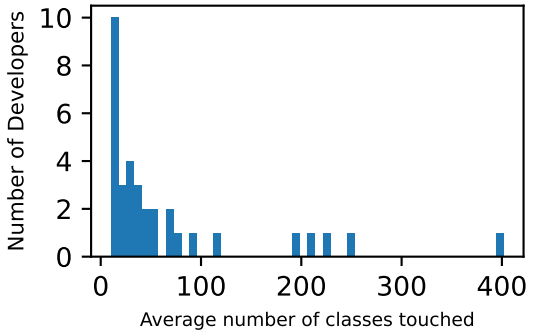
(b) Joiner (μ: 43.51, median:30, mode:13, std:49.4 and max:584)



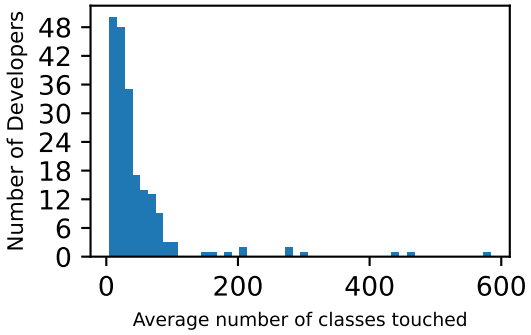
(c) Moderate founder (μ: 61.6, median:41.5, mode:7, std:87.9 and max:555)



(d) Moderate later joiner (μ: 41.95, median:30.0, mode:28, std:39.1 and max:316)

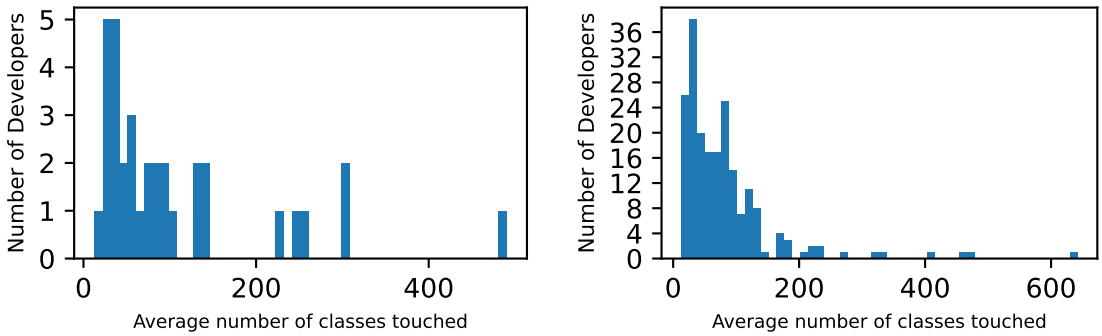


(e) Sustained founder (μ: 68.59, median:34.0, mode:40, std:87.6 and max:402)



(f) Sustained later joiner (μ: 47.98, median:29, mode:13, std:70.97 and max:584)

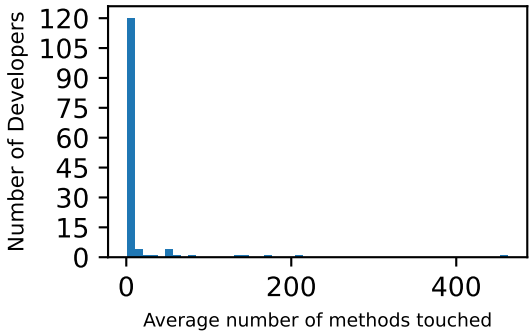
Figure 7: Classes for ten commits for different categories of founder versus different categories of late joiner



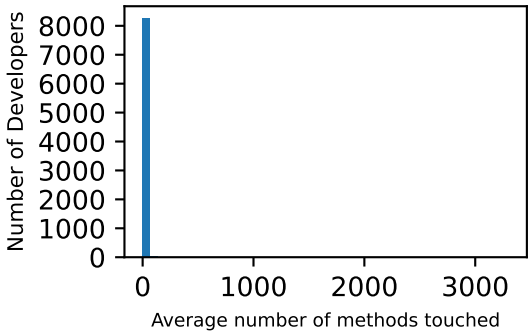
(a) Founder (μ: 107.5, median:72.5, mode:143, std:105.94 and max:490)

(b) Joiner (μ: 81.91, median:64, mode:37, std:80.41 and max:642)

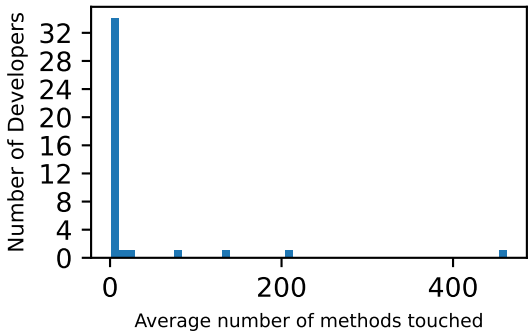
Figure 8: Classes for twenty commits for different categories of founder versus different categories of late joiner



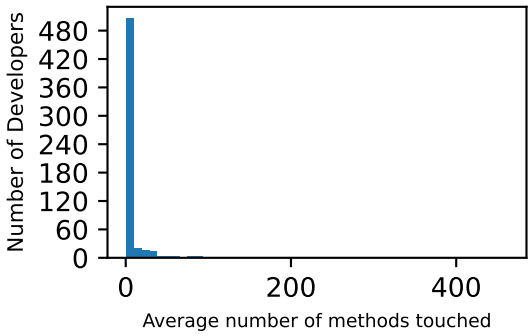
(a) Founder (μ: 13.1, median:1, mode:1, std:49.03 and max:463)



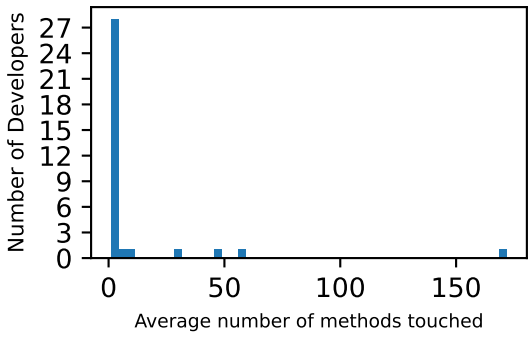
(b) Joiner (μ: 2.94, median:1, mode:1, std:39.75 and max:3301)



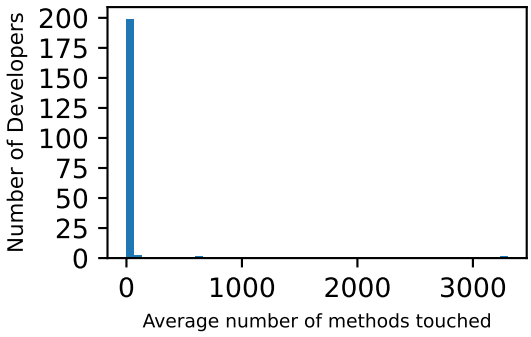
(c) Moderate founder (μ: 24.85, median:1.0, mode:1, std:81.44 and max:463)



(d) Moderate later joiner (μ: 7.82, median:1.0, mode:1, std:28.26 and max:462)

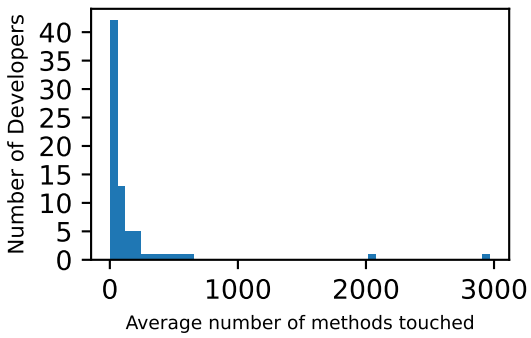


(e) Sustained founder (μ: 10.59, median:1.0, mode:1, std:31.41 and max:172)

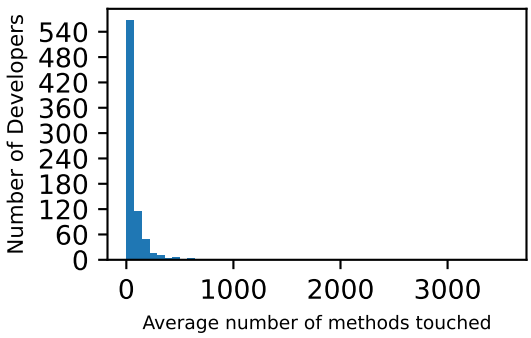


(f) Sustained later joiner (μ: 24.08, median:1, mode:1, std:235.3 and max:3301)

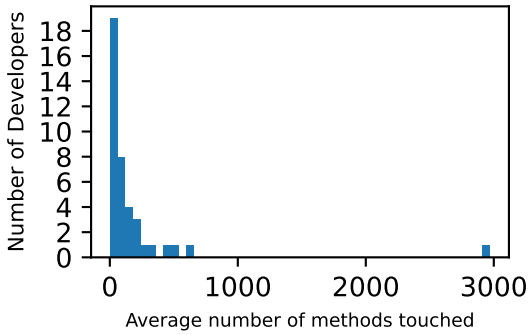
Figure 9: Methods for one commits for different categories of founder versus different categories of late joiner



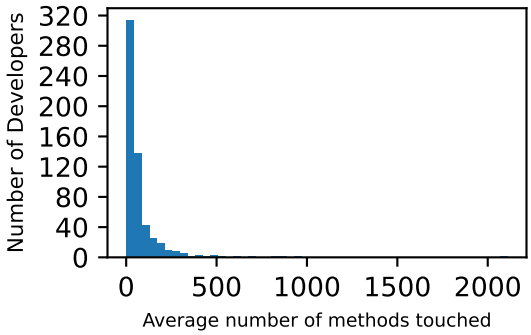
(a) Founder (μ: 164.59, median:51.0, mode:10, std:423.53 and max:2970)



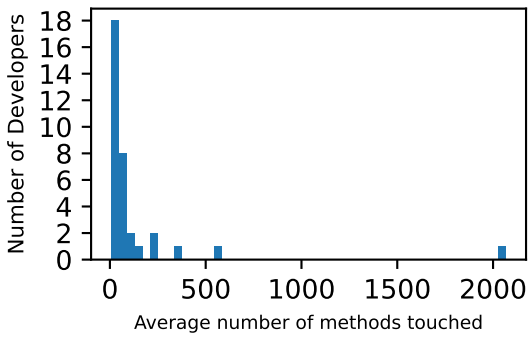
(b) Joiner (μ: 86.68, median:38, mode:15, std:207.93 and max:3559)



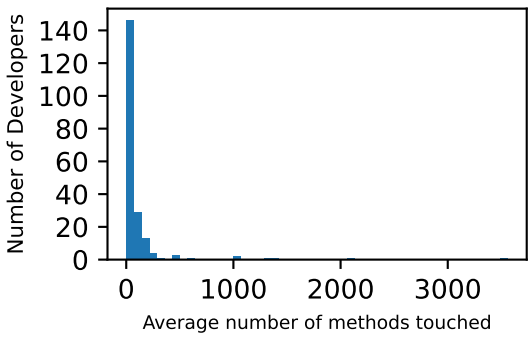
(c) Moderate founder (μ: 186.75, median:66.5, mode:10, std:474.01 and max:2970)



(d) Moderate later joiner (μ: 76.85, median:39.0, mode:10, std:138.32 and max:2109)

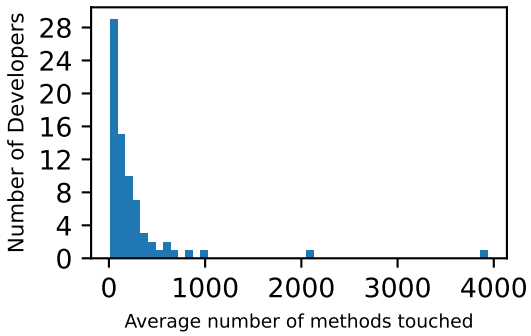


(e) Sustained founder (μ: 138.53, median:42.0, mode:15, std:360.53 and max:2069)

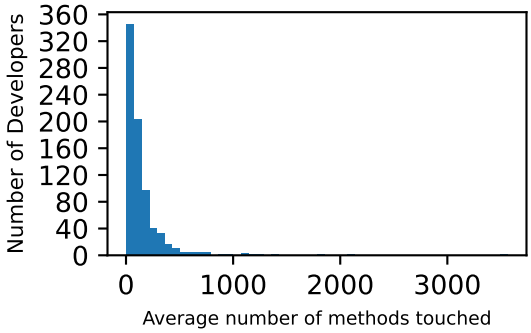


(f) Sustained later joiner (μ: 114.77, median:36, mode:15, std:333.88 and max:3559)

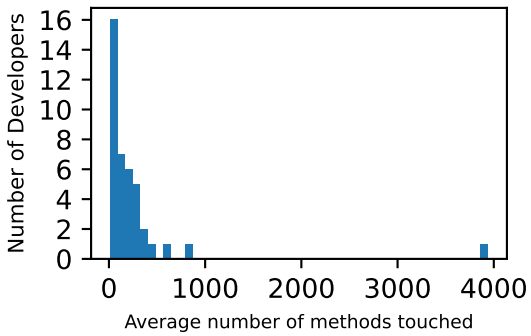
Figure 10: Methods for five commits for different categories of founder versus different categories of late joiner



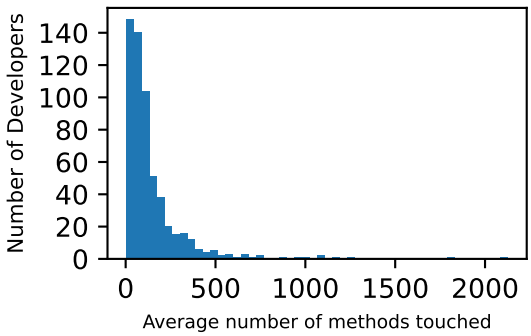
(a) Founder (μ: 263.26, median:128.0, mode:22, std:526.97 and max:3940)



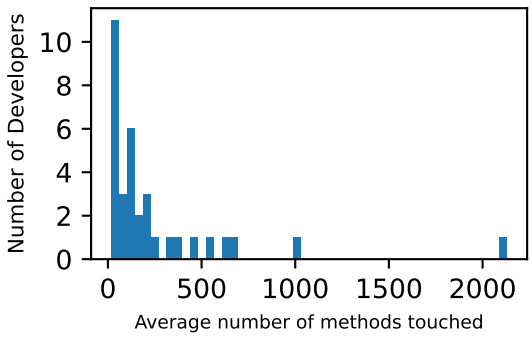
(b) Joiner (μ: 154.41, median:89, mode:18, std:244.77 and max:3561)



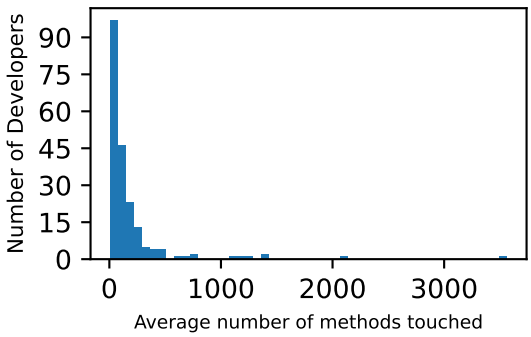
(c) Moderate founder (μ: 271.43, median:149.5, mode:208, std:619.25 and max:3940)



(d) Moderate later joiner (μ: 145.21, median:90.0, mode:18, std:193.3 and max:2126)

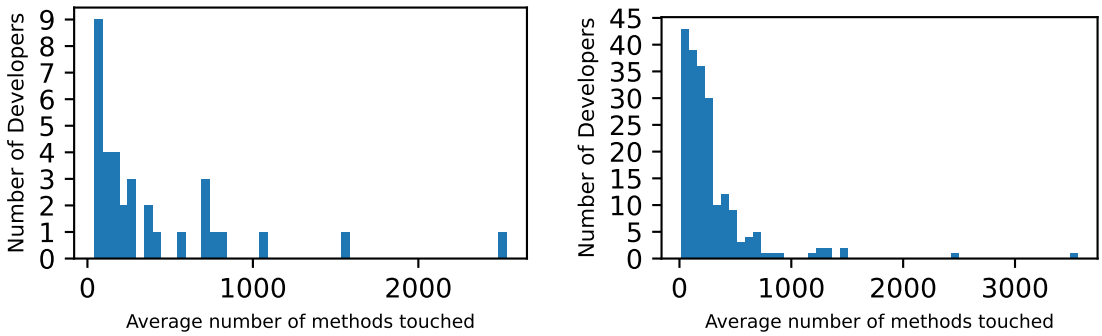


(e) Sustained founder (μ: 253.65, median:115.5, mode:22, std:401.16 and max:2132)



(f) Sustained later joiner (μ: 180.7, median:85, mode:18, std:351.99 and max:3561)

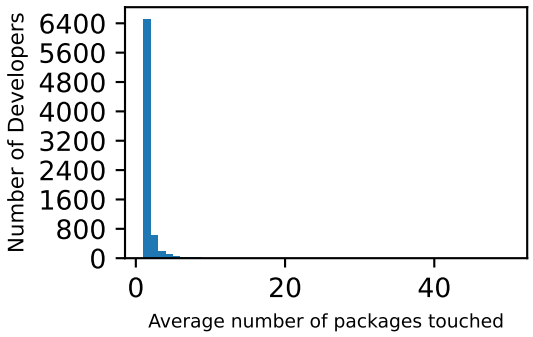
Figure 11: Methods for ten commits for different categories of founder versus different categories of late joiner



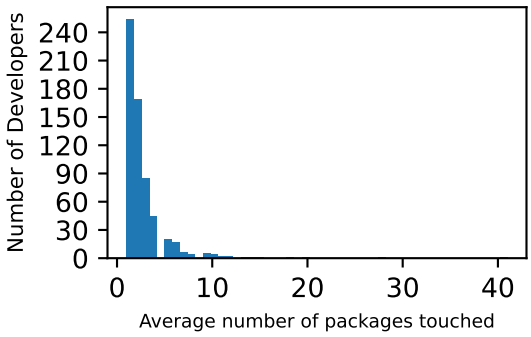
(a) Founder (μ: 401.24, median:183.5, mode:712, std:510.55 and max:2532)

(b) Joiner (μ: 295.1, median:190, mode:87, std:381.2 and max:3561)

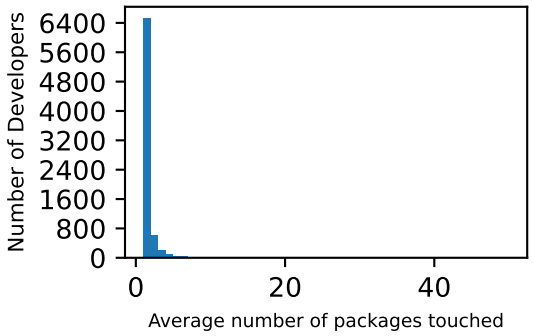
Figure 12: Methods for twenty commits for different categories of founder versus different categories of late joiner



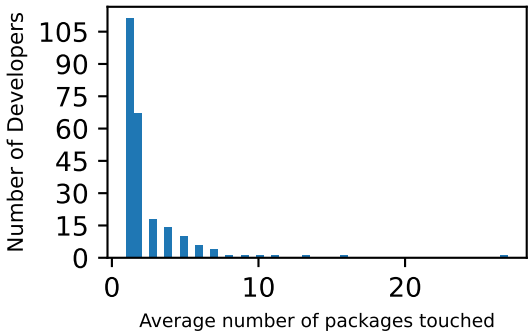
(a) Transient (μ: 1.32, median:1, mode:1, std:1.38 and max:50)



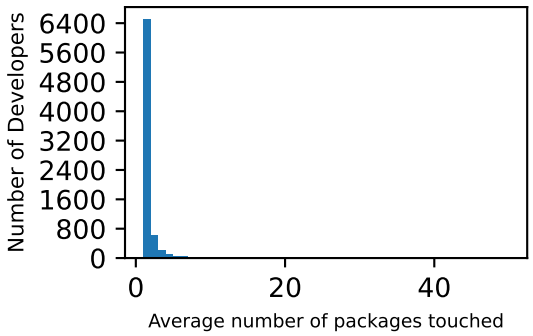
(b) Moderate (μ: 2.55, median:2.0, mode:1, std:2.88 and max:41)



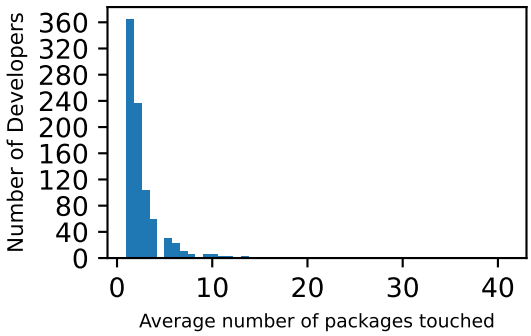
(c) Transient (μ: 1.32, median:1, mode:1, std:1.38 and max:50)



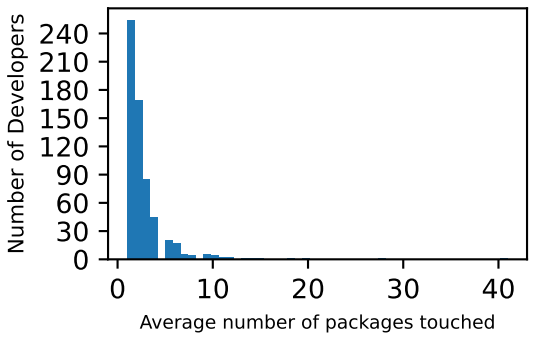
(d) Sustained (μ: 2.38, median:2, mode:1, std:2.61 and max:27)



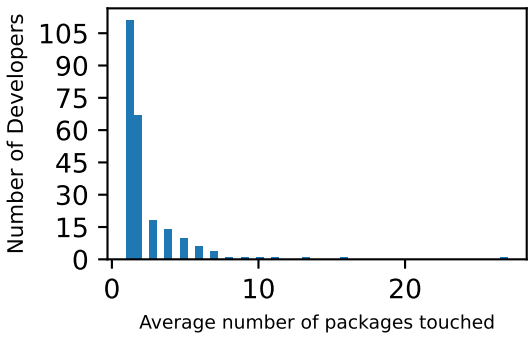
(e) Transient (μ: 1.32, median:1, mode:1, std:1.38 and max:50)



(f) Moderate sustained (μ: 2.5, median:2, mode:1, std:2.81 and max:41)

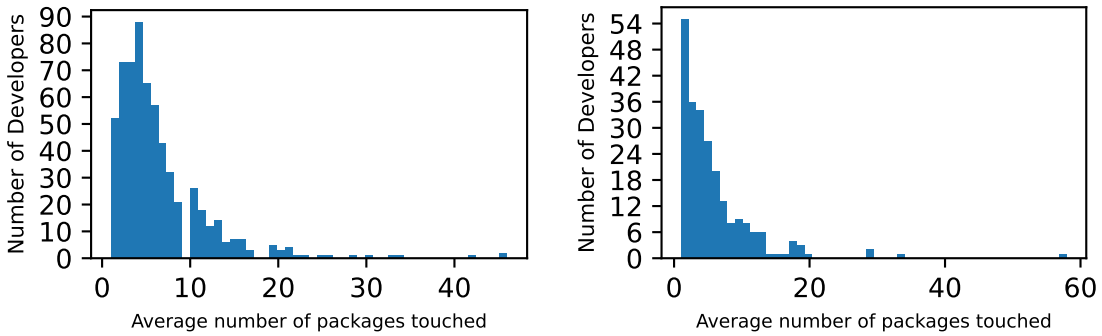


(g) Moderate (μ: 2.55, median:2.0, mode:1, std:2.88 and max:41)



(h) Sustained (μ: 2.38, median:2, mode:1, std:2.61 and max:27)

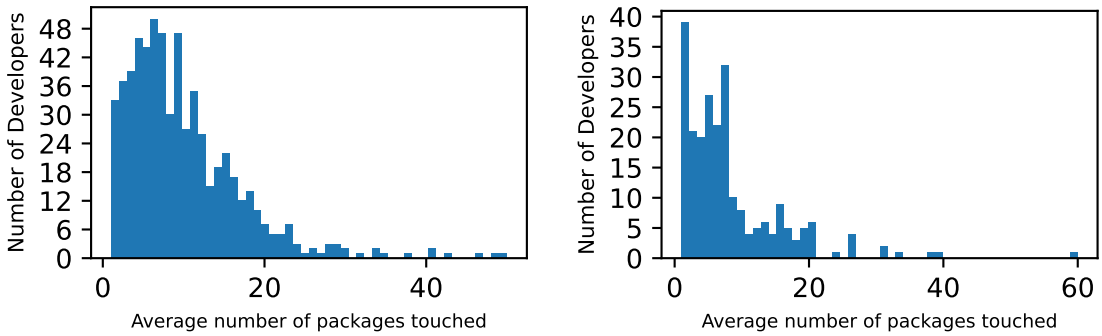
Figure 13: Packages for one commits for sample A category versus sample B category



(a) Moderate (μ: 6.31, median:5.0, mode:4, std:5.47 and max:46)

(b) Sustained (μ: 5.93, median:4, mode:3, std:5.99 and max:58)

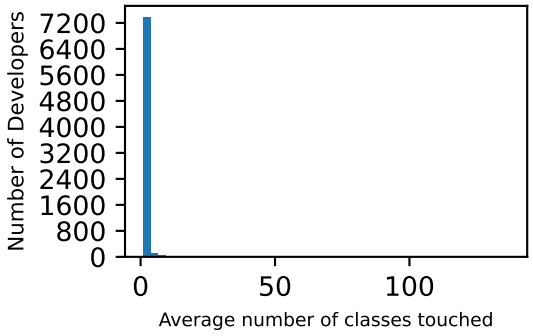
Figure 14: Packages for five commits for sample A category versus sample B category



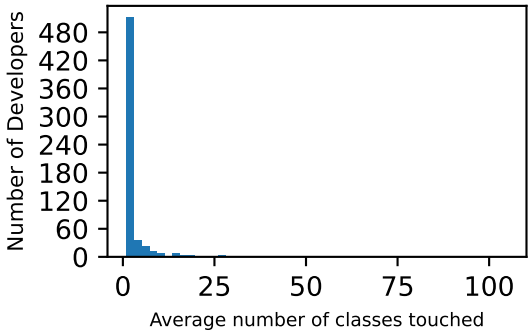
(a) Moderate (μ: 9.62, median:8.0, mode:6, std:7.33 and max:50)

(b) Sustained (μ: 8.36, median:6, mode:5, std:7.67 and max:60)

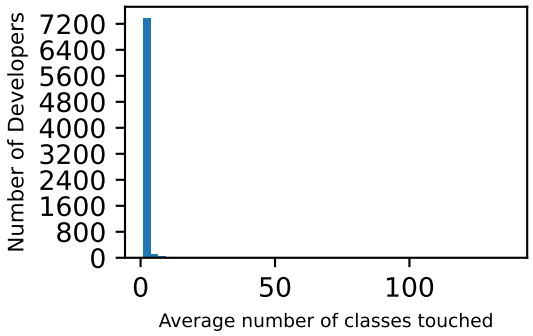
Figure 15: Packages for ten commits for sample A category versus sample B category



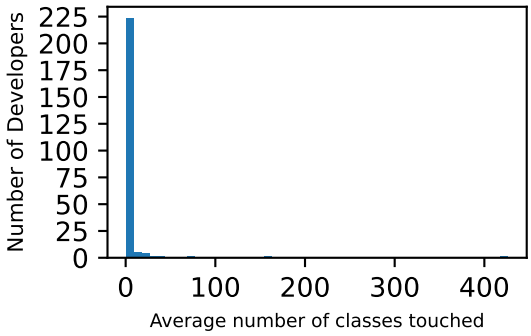
(a) Transient (μ: 1.32, median:1, mode:1, std:2.82 and max:137)



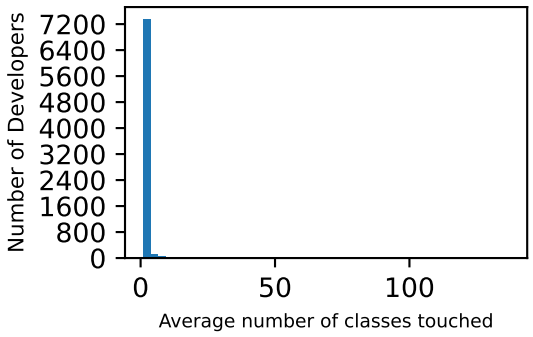
(b) Moderate (μ: 3.3, median:1.0, mode:1, std:8.85 and max:105)



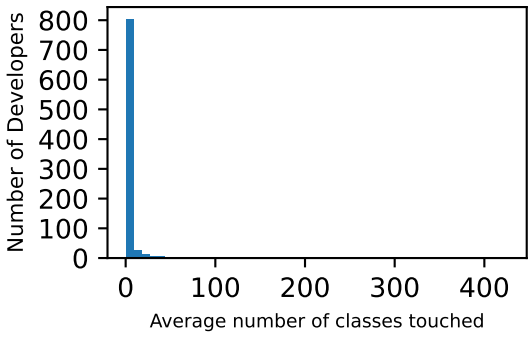
(c) Transient (μ: 1.32, median:1, mode:1, std:2.82 and max:137)



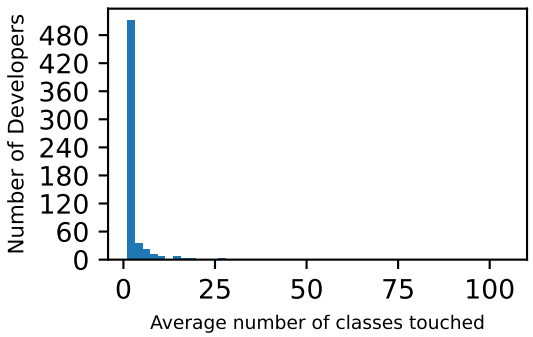
(d) Sustained (μ: 5.19, median:1, mode:1, std:30.1 and max:426)



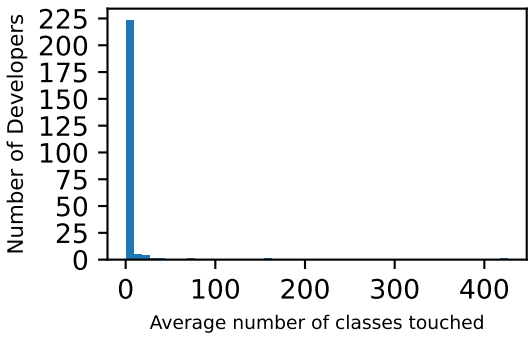
(e) Transient (μ: 1.32, median:1, mode:1, std:2.82 and max:137)



(f) Moderate sustained (μ: 3.83, median:1, mode:1, std:17.53 and max:426)

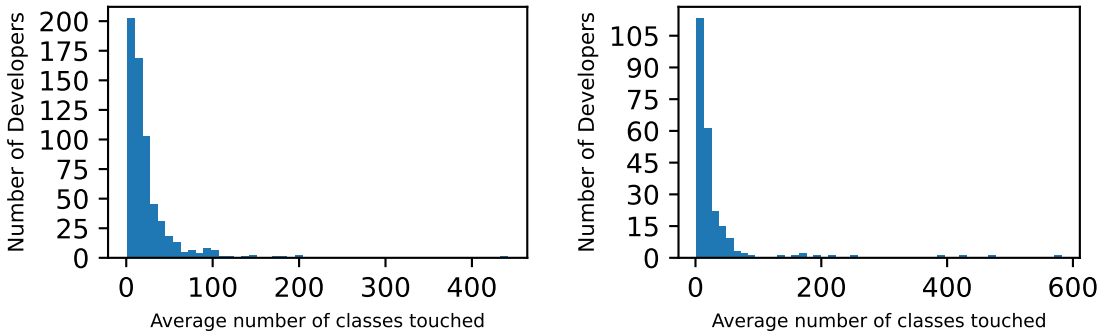


(g) Moderate (μ: 3.3, median:1.0, mode:1, std:8.85 and max:105)



(h) Sustained (μ: 5.19, median:1, mode:1, std:30.1 and max:426)

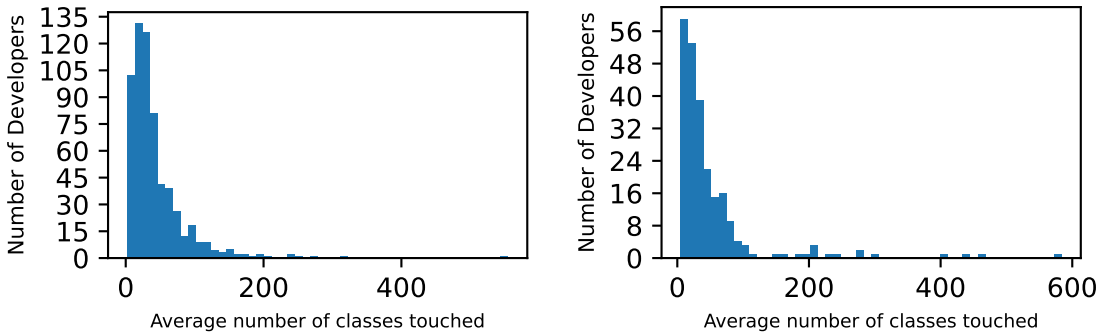
Figure 16: Classes for one commits for sample A category versus sample B category



(a) Moderate (μ: 23.29, median:14.5, mode:7, std:30.82 and max:442)

(b) Sustained (μ: 30.92, median:14, mode:9, std:66.72 and max:582)

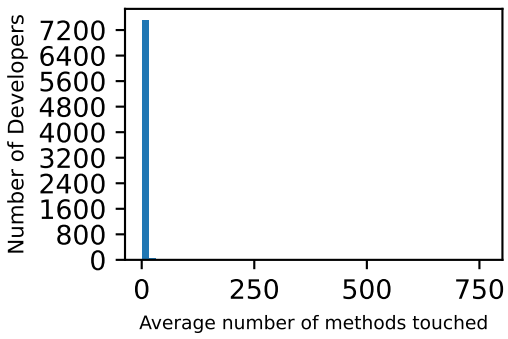
Figure 17: Classes for five commits for sample A category versus sample B category



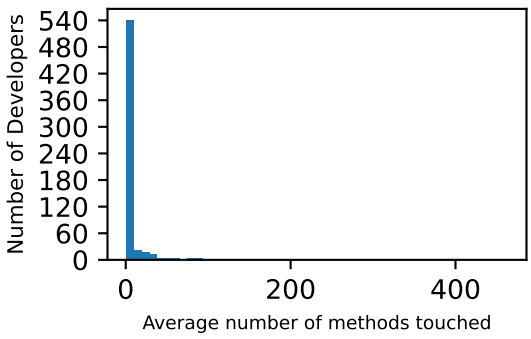
(a) Moderate (μ: 43.22, median:31.0, mode:28, std:44.05 and max:555)

(b) Sustained (μ: 50.94, median:29, mode:13, std:73.73 and max:584)

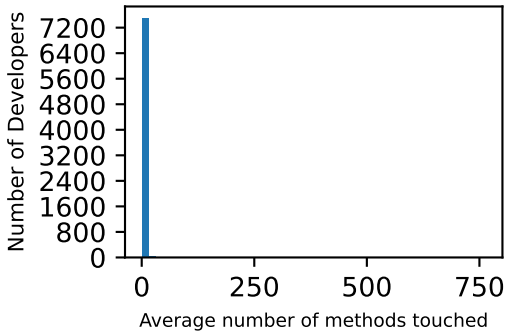
Figure 18: Classes for ten commits for sample A category versus sample B category



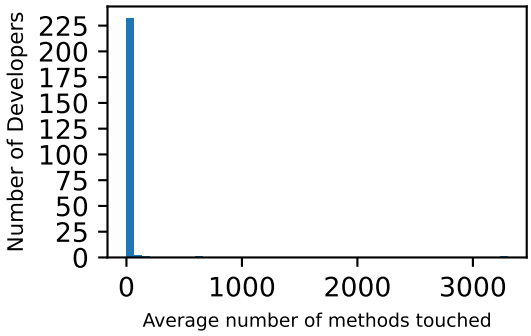
(a) Transient (μ: 2.04, median:1, mode:1, std:13.54 and max:763)



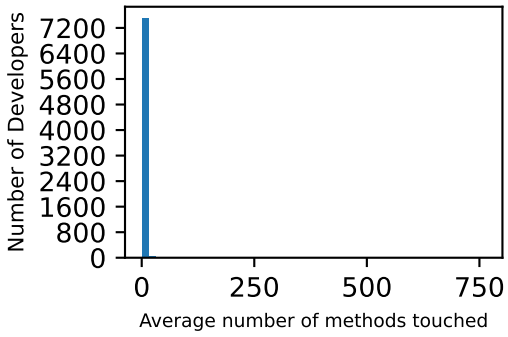
(b) Moderate (μ: 8.92, median:1.0, mode:1, std:34.39 and max:463)



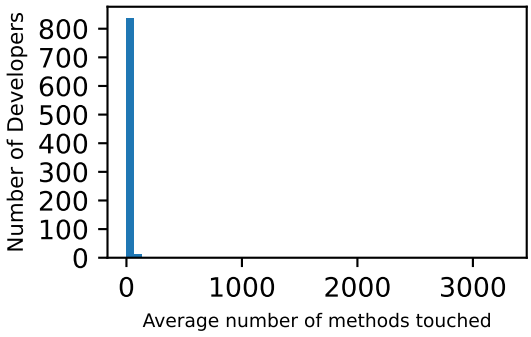
(c) Transient (μ: 2.04, median:1, mode:1, std:13.54 and max:763)



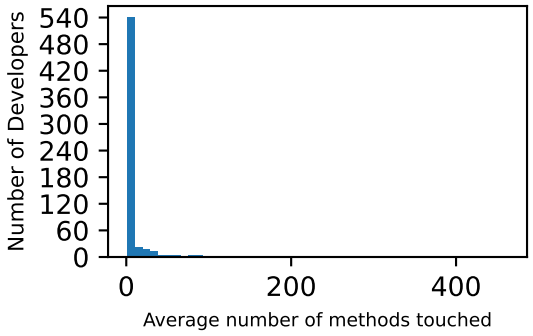
(d) Sustained (μ: 22.14, median:1, mode:1, std:218.06 and max:3301)



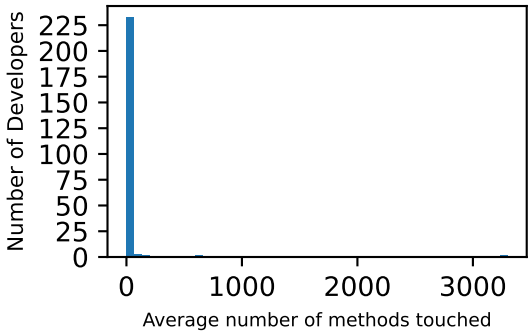
(e) Transient (μ: 2.04, median:1, mode:1, std:13.54 and max:763)



(f) Moderate sustained (μ: 12.58, median:1, mode:1, std:118.32 and max:3301)

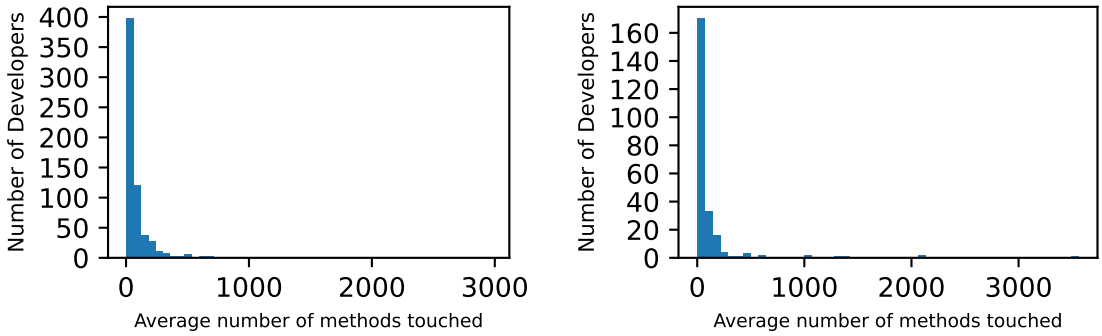


(g) Moderate (μ: 8.92, median:1.0, mode:1, std:34.39 and max:463)



(h) Sustained (μ: 22.14, median:1, mode:1, std:218.06 and max:3301)

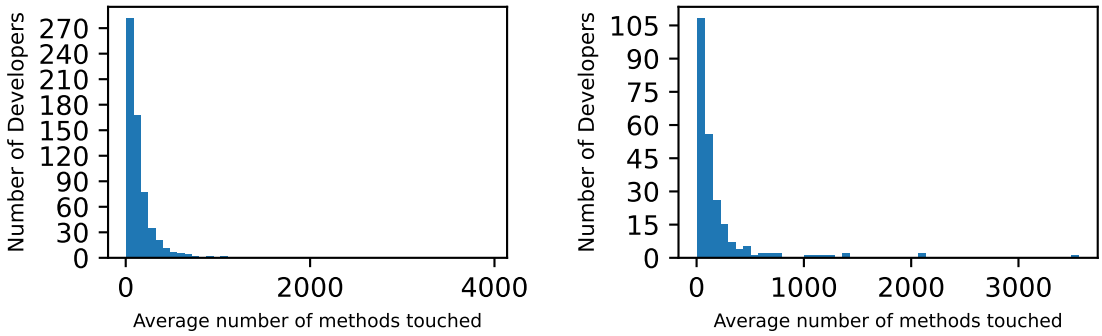
Figure 19: Methods for one commits for sample A category versus sample B category



(a) Moderate (μ: 83.94, median:40.0, mode:10, std:181.06 and max:2970)

(b) Sustained (μ: 118.18, median:37, mode:15, std:337.13 and max:3559)

Figure 20: Methods for five commits for sample A category versus sample B category



(a) Moderate (μ: 153.35, median:91.0, mode:18, std:245.1 and max:3940)

(b) Sustained (μ: 191.16, median:92, mode:18, std:359.45 and max:3561)

Figure 21: Methods for ten commits for sample A category versus sample B category